6. MASS AND BALANCE

No change in the basic Flight Manual data.

7. SYSTEM DESCRIPTION

No change in the basic Flight Manual data.

8. HANDLING, SERVICING, AND MAINTENANCE

No change in the basic Flight Manual data.

9. OPERATIONAL INFORMATION

9.1 TRAINING

The following is intended for use when performing Special Takeoff and Landing Operations training.

9.1.1 Engine and Transmission Power Training Limits

Engines: Allison 250 - C 20 B

Transmission: ZF FS 72E

NOTE The following limits are for OEI-training purposes only.

CONDITION	TRANSMISSION (HELICOPTER) LIMITS MAX TORQUE %	ENGINE OPERATING LIMITS		
		MAX TOT °C	MAX N_1 %	MAX N_2 %
One Engine Inoperative				
Max Continuous Power	86	779	105	102
30-min Power	86	800	105	102
2.5-min Power	95	800	105	102
Transient (Max 6 sec above 800 °C)		843		
Transient (Max 16 sec)	110			

9.1.2 OEI Training Procedures

NOTE For training purposes, OEI emergency procedures will normally be simulated with both engines operating.

9.1.2.1 OEI Simulation with All Engines Operating

To simulate OEI operation using both engines at reduced power, follow the emergency procedures of the respective type of operation in conjunction with the corresponding max takeoff/landing gross mass.

For determining the training AEO-torque values, use the OEI Simulation 2.5-min power diagram (see Fig. 7). Transient torque limits as required, however, not more than Takeoff Power.

9.1.2.2 OEI Simulation with One Engine at IDLE

For determining the training takeoff and landing gross mass in order to simulate OEI conditions during FMS 11-4 operations with one engine at IDLE, refer to "Training" paragraph applicable for the respective type of operation.

EXAMPLE: (see Fig. 7)

Determine: **2.5-min AEO-torque when simulating OEI takeoffs and landings**

Known:	OAT	15 °C
	Pressure Altitude	SL

Solution:
1. Enter chart at known OAT (15 °C).
2. Move up to known pressure altitude (SL).
3. Move right and read indicated torque = 54.5 %.

NOTE Resultant torque value is the 2.5-min power for simulating OEI conditions when using an FMS 11-4 type of operation Maximum Takeoff or Landing Gross Mass.

OEI-SIMULATION 2.5-MIN POWER

ALLISON 250-C20B

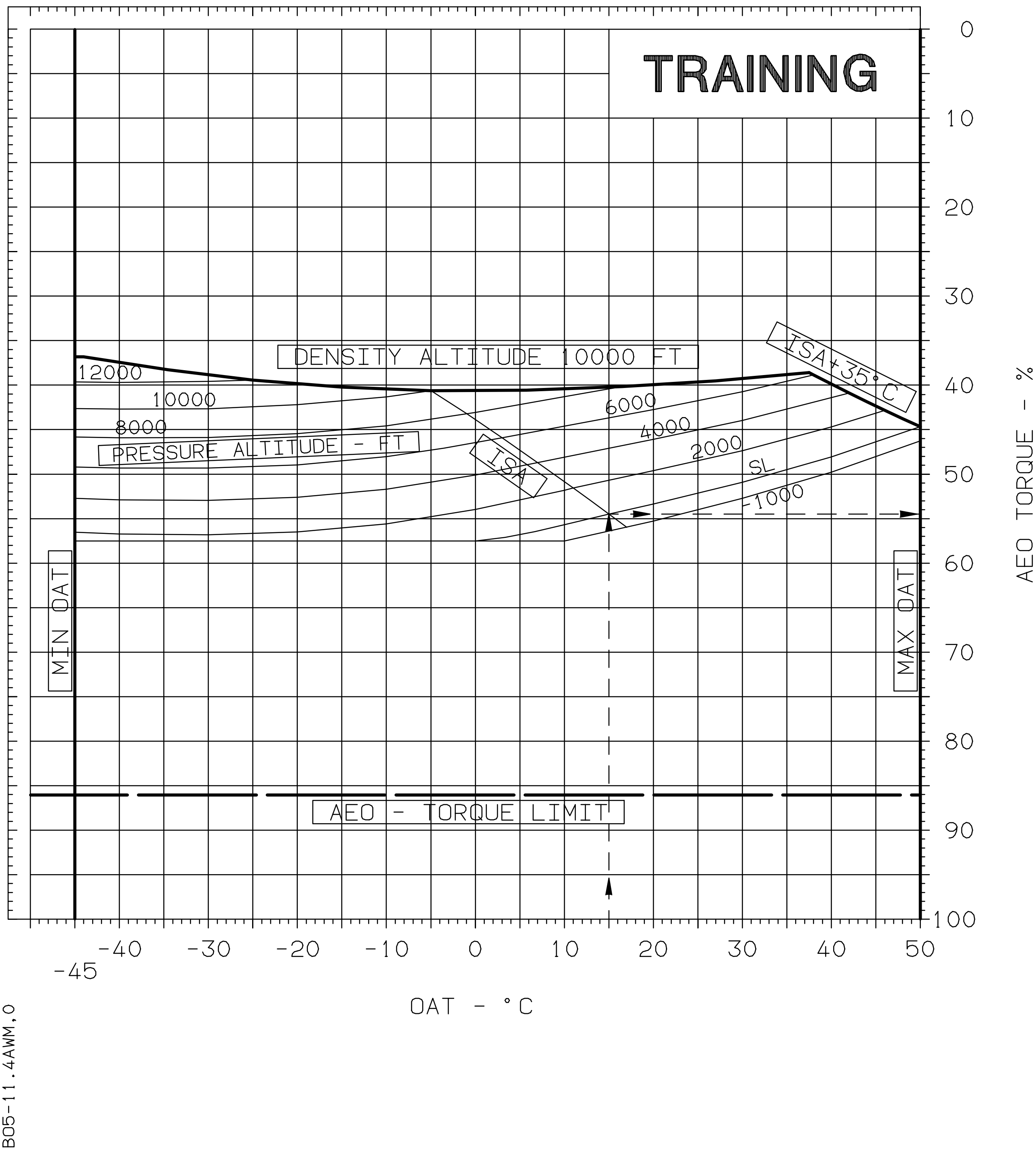


Fig. 7 OEI-simulation 2.5-min Power

9.2 OEI OUT-OF-GROUND-EFFECT HOVER CEILING

The OEI Hover Ceiling chart (see Fig. 8) has been established for flight planning purposes only and is to be used in calculating OEI hover gross mass operational performance requirements.

NOTE OEI hover with tailwind component shall be avoided.

<u>EXAMPLE</u>: (see Fig. 8)

Determine: **Max gross mass for OEI hover, wind calm**

Known:	OAT	15 °C
	Pressure altitude	SL
	Wind	Calm

Solution:
1. Enter chart at known OAT (15 °C).
2. Move up to known pressure altitude (SL).
3. Move right and read gross mass = <u>1675 kg</u>.

Determine: **Max gross mass for OEI hover, with wind**

Known:	OAT	15 °C
	Pressure altitude	SL
	Headwind	10 kt

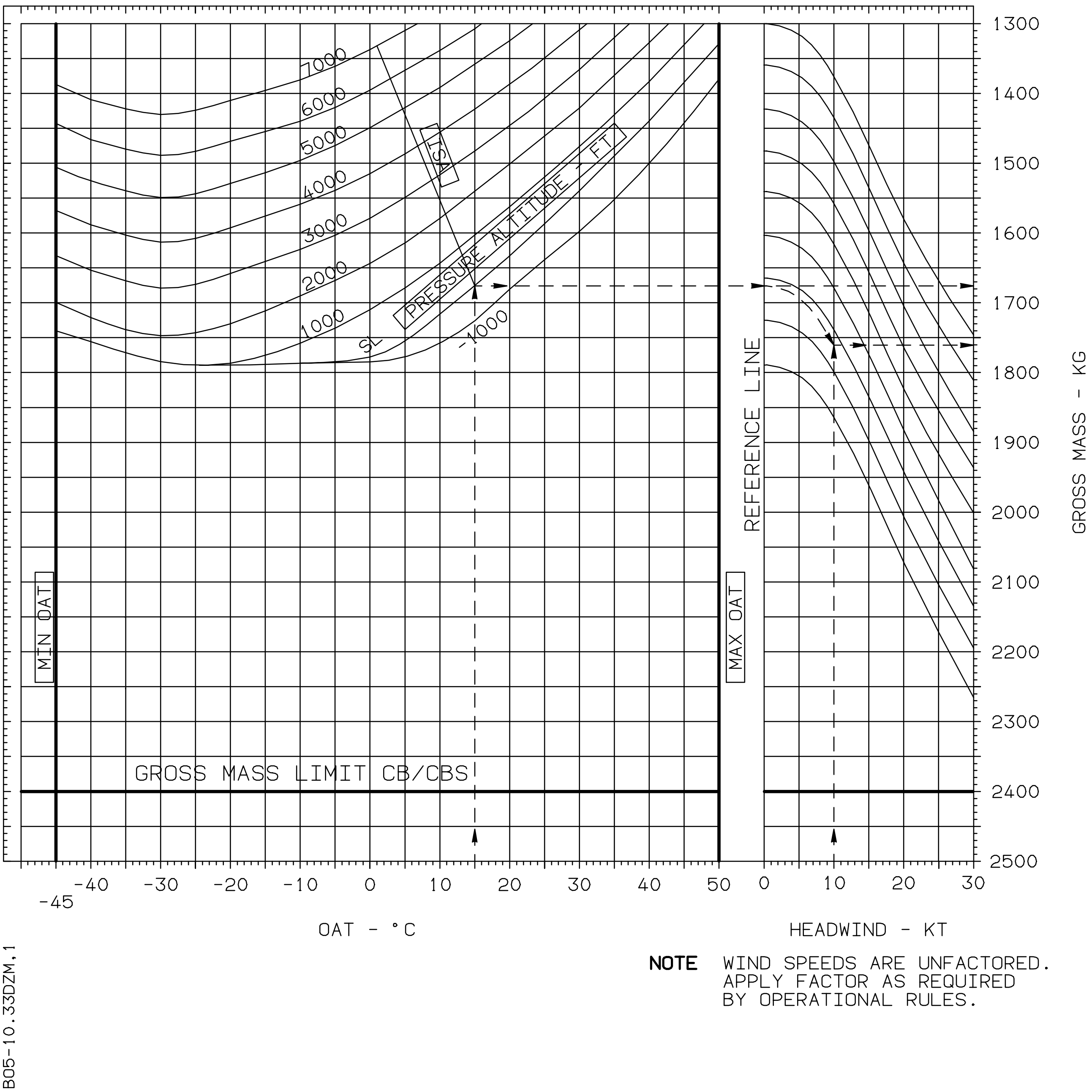
Solution:
1. Enter chart at known OAT (15 °C).
2. Move up to known pressure altitude (SL).
3. Move right to reference line and then down following curve of guide lines to known headwind (10 kt).
4. Move right and read gross mass = <u>1762 kg</u>.

OEI HOVER CEILING OUT OF GROUND EFFECT

1 X ALLISON 250-C20B

OEI – 2.5-MIN POWER (810 °C TOT, 110% TORQUE)

ANTI-ICING – OFF
BLEED AIR CONSUMERS – OFF



NOTE WIND SPEEDS ARE UNFACTORED. APPLY FACTOR AS REQUIRED BY OPERATIONAL RULES.

Fig. 8 OEI Hover Ceiling Out of Ground Effect